**Indian Institute of Science Education and Research (IISER) Mohali**

*Sector 81, Knowledge City, S A S Nagar, Mohali, Punjab-140 036*

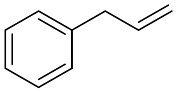
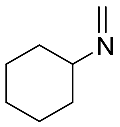
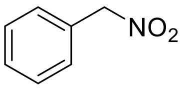
# CHM 102 Basic Organic Chemistry

## Practice Problem Set - 1 – Spring 2024

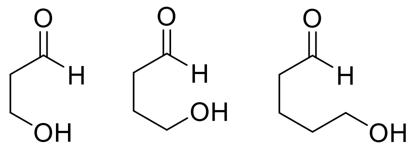
**Q1.** Draw all possible resonance structures for the following compounds:

(a) $O_3$ (b) $N_2O$ (c) $HN_3$ (d) HNCO

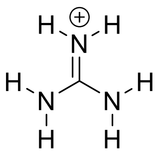
**Q2**. Draw the important tautomeric structures for the following compounds. Indicate the most stable structure with reasons and position of the equilibrium.

(a)  (b)  (c) 

**Q3**. Arrange the equilibrium constants for the following analogous compounds (hydroxy alkylcarboxaldehydes) towards their respective ring-closure tautomeric structures. Also, indicate the reason. (Draw the structures with equilibrium).



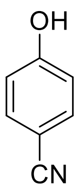
**Q4**. Why the protonated guanidium ion has all three C-N bond distances identical?



**Q5**. Draw the following MOs

(a) HOMO of 1,3-butadiene (b) LUMO of allyl cation (c) HOMO of 1,3,5-hexatriene

**Q6**. The $pK_a$ of 1 and 3 are nearly the same, but 2 has lower pKa value than that of 4. Why?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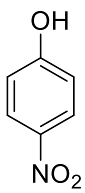
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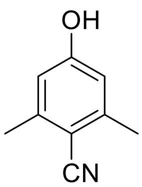
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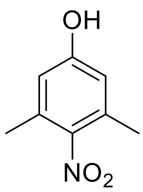

4

**Q7**. Formaldehyde has a fairly large dipole moment of 2.33 D, but CO has a small dipole moment of 0.11 D. Use resonance and electronegativity arguments to explain these results.

**Q8**. Explain why phenol is more acidic (pKa = 10) than cyclohexanol (pKa = 16).

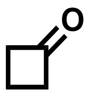
**Q9**. Predict the possible dimeric interaction mode in $CH_3Cl$. (Hint: Use the bond dipole(s)/molecular dipole moment for prediction of the charge centres)

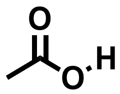
**Q10**. Nitromethane (pKa = 10) is more acidic than methane (pKa = 16). Why?

**Q2.** Draw the structure for compounds having IUPAC names as follows:

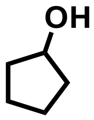
(a) 2-Ethyl-4-methyl-3-vinylpent-3-en-1-ol
(b) 3-Allyl-3-ethyl-1-methylcyclohexan-1-ol
(c) 1,2,3,4-Tetrahydropyrazine
(d) 4-(Pent-4-en-2-yn-1-yl)cyclohex-1-ene
(e) Thietan-2-one
(f) 4-Chloro-5-fluoro-1,2-oxazole

**Q3.** Draw the significant bond dipoles and molecular dipoles for the following molecules:



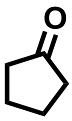




**Q4.** Identify the trend in the boiling points among the following compounds and indicate the reason.

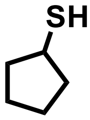



A

B




C




D

**Q5.** Draw the frontier molecular orbitals for the following molecules:

(a) Cyclobutadiene
(b) 1,3,5,7-Octatetraene